

Name:		Centre/Index Number:		Class:	
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DUNMAN HIGH SCHOOL

Preliminary Examination

Year 6

H2 BIOLOGY

Paper 4 Practical

9744/04

31 August 2021

2 hours 30 minutes

Candidates answer on the Question Paper

READ THESE INSTRUCTIONS FIRST

Write your centre number, index number, name and class at the top of this page.

Give details of the practical shift and laboratory where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the question paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	28
2	27
Total	55

This document consists of **18** printed pages (including this cover page) and **0** blank page.

- 1 Plant cells contain enzymes that catalyse metabolic reactions. One of these enzymes, catalase, catalyses the release of oxygen from the substrate hydrogen peroxide (H_2O_2). The equation for this reaction is:



A plant extract solution containing catalase can be prepared.

When hydrogen peroxide solution and a plant extract solution containing catalase are mixed, oxygen is released. The released oxygen forms bubbles that rise to the surface and make foam on the surface of the solution.

In this investigation, you will compare the activity of catalase at different temperatures by measuring the height of the foam produced by the release of oxygen.

You are required to test the activity of catalase in a plant extract solution, **E**, using a suitable number of temperatures, including a maximum temperature of 50°C.

You are provided with:

- a beaker, labelled **hot water**, in which to collect hot water at a temperature of 50°C to 55°C
- a beaker, labelled **tap water**, in which to collect tap water
- a mixture of ice and water, in a beaker labelled **ice water**
- plant extract solution **E**, in a beaker labelled **E**
- 1.0% hydrogen peroxide solution **S**, in a beaker labelled **S**

Plant extract solution E is an irritant. Suitable eye protection should be worn. If E comes into contact with your skin, wash it off immediately under tap water. Care should be taken when using hot water.

- (a) (i) State the temperatures that you will use in your investigation.

[1]

Proceed as follows.

- 1** Label the required number of test-tubes with the temperatures you decided in **(a)(i)**.
- 2** Take up 5.0 cm³ of plant extract solution **E** into a syringe.
- 3** Position the syringe so that the nozzle touches the side of one of the labelled test-tubes.
- 4** Gently push the plunger of the syringe so that **E** runs slowly down the side of the test-tube to the bottom. This is shown in Fig. 1.1.

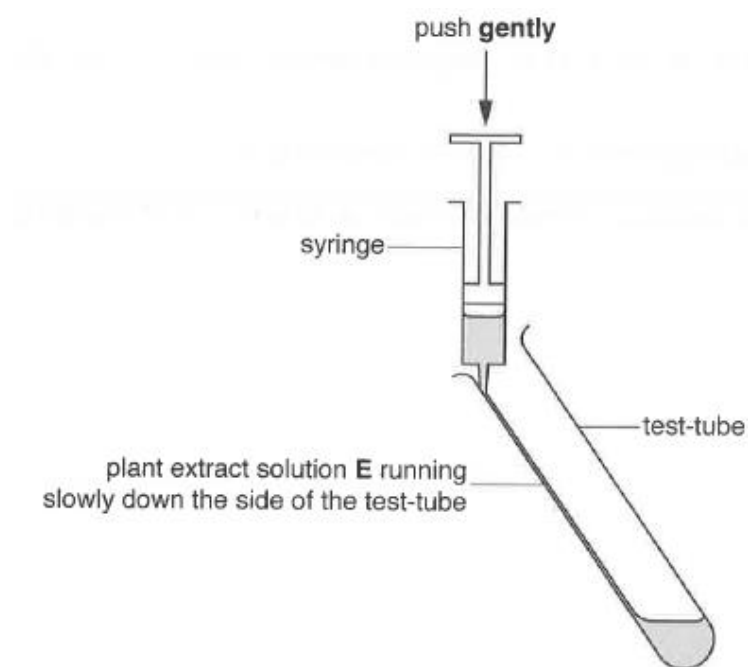


Fig. 1.1

- 5** Put this test-tube into the test-tube rack.
- 6** Repeat steps **2-5** for each of the other test-tubes that you labelled in step **1**.
- 7** Label beakers with each of the temperatures that you will be using in your investigation.

Before proceeding further, use the beaker labelled **hot water** to collect approximately 400 cm³ of hot water from where it is provided in the laboratory.

The beaker labelled **tap water** can be filled as necessary from the tap.

- 8** Set up water-baths at the required temperatures in the beakers labelled in step 7, using **hot water**, **tap water** and **ice water**. For the water-bath at 50°C, you may need to carefully heat the water using a Bunsen burner, so place this water-bath on a tripod and gauze. Sufficient water should be used to ensure that the contents of the test-tubes will be covered in step **9**, but volumes of water in different water-baths do not need to be the same. There should be room to add additional water to adjust the temperatures of the water-baths in step **10**, as necessary.
- 9** Put each of the labelled test-tubes into the corresponding water-bath. Leave the test-tubes for five minutes.
- 10** Starting two minutes into this five-minute period, adjust the temperature of each water-bath to the required temperature once by adding **hot water**, **tap water** or **ice water**, as necessary. For the water-bath at 50°C, you will need to carefully heat the water using a Bunsen burner.
- 11** After five minutes, carefully add 2.0 cm³ of **S** down the side of each labelled test-tube in the same way as that shown for **E** in Fig. 1.1. Mix gently.
- 12** Place each labelled test-tube back into the appropriately labelled water-bath. Leave the test-tubes for another five minutes. At the start of this five-minute period, adjust the temperatures of the water-baths as described in step **10**. After this adjustment, do not make any further adjustments to the temperatures of the water-baths.
- 13** After five minutes, take the test-tubes out of the water-baths and put them into the test-tube rack.
- 14** Measure the height of the foam in each of the test-tubes.
 - (ii) Use the space provided to record your results from step **14** in a suitable format.

- (iii) From your results, state a valid conclusion about the optimum temperature of catalase in **E**.

[1]

- (iv) Using the height of foam to measure oxygen production is one significant source of error in this procedure.

State **one other** significant source of error in this procedure to investigate the effect of temperature on catalase activity.

Suggest **one** modification to the procedure that would reduce this source of error.

Source of error _____

Modification _____

[2]

- (v) Describe a method, using suitable apparatus, to measure oxygen production more accurately in this investigation.

You may use a diagram to help your description.

[2]

- (b) A student investigated the activity of catalase at room temperature in extracts from two species of plants, **A** and **B**. The activity of catalase was measured in ten samples of extract **A** and ten samples of extract **B**. All other variables were kept constant.

Table 1.1 shows the results of the investigation.

Table 1.1

sample number	activity of catalase / arbitrary units	
	extract A	extract B
1	7.5	9.8
2	9.4	8.5
3	7.9	8.4
4	7.6	7.8
5	8.4	10.1
6	8.4	10.5
7	8.5	11.4
8	7.8	9.2
9	7.6	11.2
10	8.6	8.8
mean ($\bar{x}$)	8.2	9.6
standard deviation (s)	0.60	1.23
variance (s^2)		

- (i) Complete Table 1.1 by calculating the variance (s^2) for the activity of catalase in each of the two extracts: extract **A** and extract **B**.

[1]

- (ii) A t -test can be used to determine whether there is any significant difference between the activity of catalase in extract **A** and the activity of catalase in extract **B**.

Calculate the value of t and the number of degrees of freedom, using these formulae:

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \quad \nu = n_1 + n_2 - 2$$

key to symbols

s = standard deviation

$\bar{x}$ = mean

n = sample size (number of observations)

ν = degrees of freedom

Show your working.

Value of t = _____

Number of degrees of freedom = _____

[2]

(iii) For this t -test, the student proposed the null hypothesis:

there is no difference between the activity of catalase from species **A** and the activity of catalase from species **B**.

Table 1.2 shows the critical values for t at several different probabilities and degrees of freedom.

Table 1.2

degrees of freedom	probability, p				
	0.5	0.1	0.05	0.01	0.001
1	1.00	6.31	12.71	63.66	636.62
2	0.82	2.92	4.30	9.92	31.60
3	0.76	2.35	3.18	5.84	12.92
4	0.74	2.13	2.78	4.60	8.61
5	0.73	2.02	2.57	4.03	6.87
6	0.72	1.94	2.45	3.71	5.96
7	0.71	1.89	2.36	3.50	5.41
8	0.71	1.86	2.31	3.36	5.04
9	0.70	1.83	2.26	3.25	4.78
10	0.70	1.81	2.23	3.17	4.59
11	0.70	1.80	2.20	3.11	4.44
12	0.70	1.78	2.18	3.05	4.32
13	0.69	1.77	2.16	3.01	4.22
14	0.69	1.76	2.14	2.98	4.14
15	0.69	1.75	2.13	2.95	4.07
16	0.69	1.75	2.12	2.92	4.01
17	0.69	1.74	2.11	2.90	3.97
18	0.69	1.73	2.10	2.88	3.92
19	0.69	1.73	2.09	2.86	3.88
20	0.69	1.72	2.09	2.85	3.85

Use Table 1.2 and your answers to **(b)(ii)** to decide whether the null hypothesis suggested by the student should be accepted or rejected.

Explain your answer.

accept or reject null hypothesis _____

explanation _____

_____ [1]

- (c)** A second student wanted to investigate the effect of pH on the activity of catalase in plant extract solution **E**. The student planned to measure the activity of catalase by adding hydrogen peroxide solution to **E** and measuring the height of foam produced after five minutes. All variables, other than pH, were kept the same.

- (i)** Using your biological knowledge, predict **and** explain the trend in the height of foam produced as the pH changes.

_____ [4]

- (ii)** State how the student could vary the pH.

_____ [1]

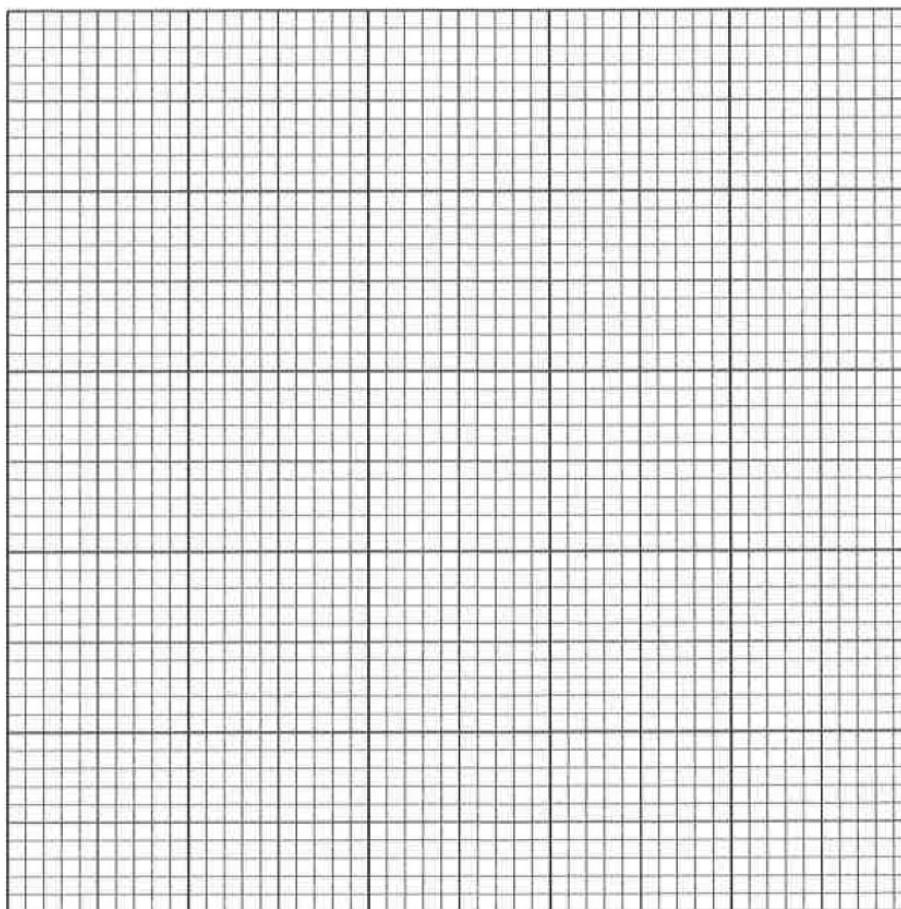
- (d) Another student investigated the effect of hydrogen peroxide concentration on the activity of catalase.

Table 1.3 shows the results of this investigation.

Table 1.3

percentage concentration of hydrogen peroxide	activity of catalase /arbitrary units
0	0.0
5	57.5
10	150.0
15	235.0
20	247.5
25	247.5

- (i) Use the grid provided to display the results shown in Table 1.3.



[4]

- (ii) Explain the change in the activity of catalase as the concentration of hydrogen peroxide increases from 15% to 25%.

[3]

- (iii) Show on your graph in (d)(i) how to estimate the activity of catalase when the concentration of hydrogen peroxide is 6.5%.

[1]

- (iv) With reference to your graph in (d)(i), state the estimated activity of catalase when the concentration of hydrogen peroxide is 6.5%

Activity of catalase = _____ arbitrary units
[1]

[Total: 28]

2 During this question you will require access to a microscope and slide **M1**.

M1 is a slide of a stained transverse section through a leaf of a land plant.

You are not expected to be familiar with this specimen.

- (a) Use your microscope to observe the different tissues in the region of slide **M1** shown by the shaded area in Fig. 2.1.

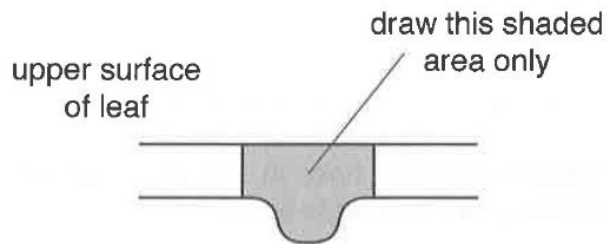


Fig. 2.1

- (i) Use the space provided to draw a large plan diagram of the part of the leaf on slide **M1** shown by the shaded area in Fig. 2.1.

A plan diagram shows the arrangement of different tissues, including their correct shapes and proportions. No cells should be drawn.

Labels are **not** required.

- (ii) Observe the outer layer of cells on the upper surface of the leaf on slide **M1**. This outer layer of cells is called the upper epidermis.

Select four cells from the upper epidermis that form a chain.

Make a large drawing of this chain of **four** cells.

Labels are **not** required.

[4]

- (iii) You can assume that the actual length of the chain of four cells that you have drawn in **(a)(ii)** is $68\mu\text{m}$.

Use this information to calculate the magnification of your drawing in **(a)(ii)**.

Show all the steps in your calculation, including the appropriate units.

Magnification = $\times$ _____
[4]

- (iv) In land plants, one of the main functions of the epidermis is to reduce water loss.

From your observation of slide **M1**, suggest **one** way in which the upper epidermis of land plants is adapted to reduce water loss.

[1]

- (b) In pondweed, reducing water loss from leaves is not necessary and the epidermis has a number of functions, including photosynthesis and gas exchange.

A student investigated the effect of different light intensities on the rate of photosynthesis of leaves of pondweed, by measuring the uptake of carbon dioxide.

The student placed pondweed in beakers of distilled water and exposed the pondweed to three different light intensities: high, medium and low. At the end of the experiment, the student removed a sample of water from each beaker and transferred it to a plastic vial. Hydrogencarbonate indicator was then added to each sample.

Hydrogencarbonate indicator changes colour depending on the concentration of carbon dioxide. The concentration of carbon dioxide in air is normally 0.04%.

- At 0.04% carbon dioxide concentration, the indicator is red.
- At less than 0.04% carbon dioxide concentration, the indicator is purple.
- As the carbon dioxide concentration increases above 0.04%, the indicator changes colour from red to orange-red and eventually yellow.

- (i) State the dependent variable in this investigation.

[1]

- (ii) Two variables in this investigation that need to be standardised are:

- the volume of the sample of water removed from each beaker and transferred to a plastic vial
- the volume of hydrogencarbonate indicator added to each sample.

Describe how **three other** variables in this investigation could be standardised.

[3]

- (iii) The student used a desk lamp with a 60W light bulb as the source of light.

Describe a method the student could use to obtain a range of different light intensities from the light source.

The method should allow the same light intensities to be obtained if repeated by another student.

[2]

You are provided with three samples of water, **P**, **Q** and **R**, in plastic vials. Each sample was taken from one of the three beakers containing pondweed after exposure to different light intensities.

You are also provided with hydrogencarbonate indicator **B**, in a beaker labelled **B**.

You are required to identify the light intensities to which the pondweeds in **P**, **Q** and **R** were exposed.

Proceed as follows.

- 1 Label three test-tubes **P**, **Q** and **R**.
- 2 Put 5.0cm³ of **P** into the test-tube labelled **P**.
- 3 Put 5.0cm³ of **Q** into the test-tube labelled **Q**.
- 4 Put 5.0cm³ of **R** into the test-tube labelled **R**.
- 5 Add 2.0cm³ of hydrogencarbonate indicator **B** to each of **P**, **Q** and **R** in the labelled test-tubes. Mix gently and observe the results.

- (iv) Record your results in a suitable format in the space below.

[2]

- (v) Using your results from **b(iv)**, complete Table 2.1 to match the light intensity, **high**, **medium** or **low**, to the correct sample.

Table 2.1

sample	light intensity
P	
Q	
R	

[1]

- (vi) Use your biological knowledge to explain how you identified **P**, **Q** and **R** in Table 2.1.

[2]

- (vii) Describe a suitable set of controls for this experiment.

[1]

- (viii) Suggest **one** improvement to this experiment, other than including controls or repeating the procedure, that would increase confidence in the results.

[1]

[Total: 27]

End of Paper



DUNMAN HIGH SCHOOL
Preliminary Exam
Year 6

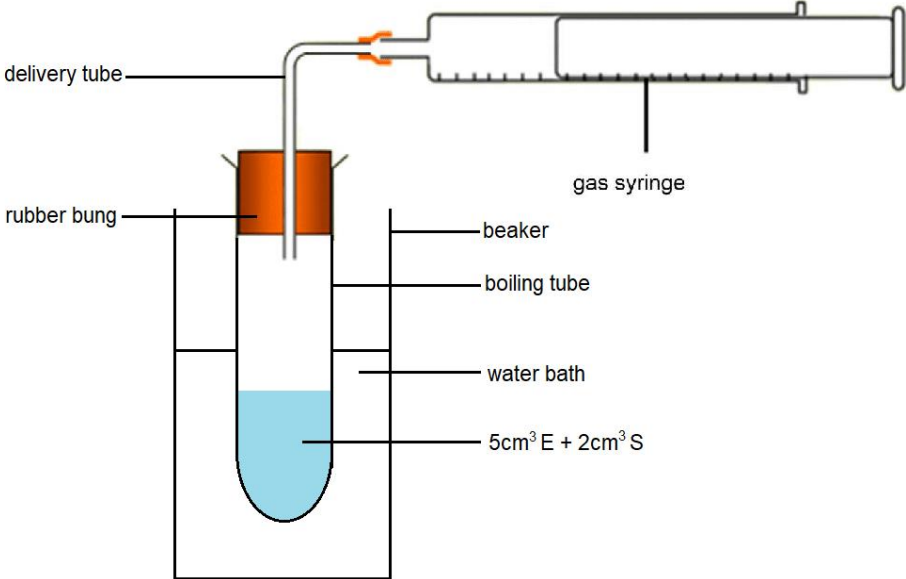
H2 Biology

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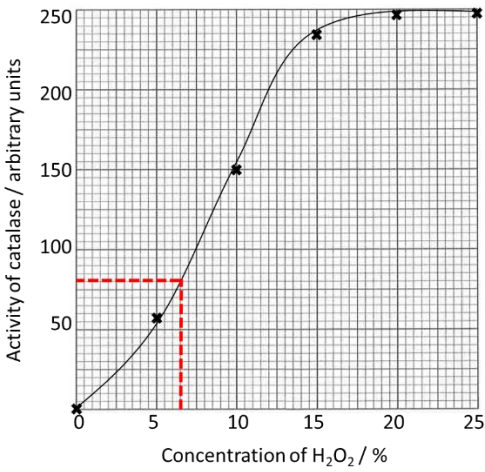
Paper 4 Practical

Answer Scheme

Question 1														
(a)(i)	Regular interval + 5 temperatures (excluding 0°C and max 50°C); Eg. 10°C, 20°C, 30°C, 40°C, 50°C	[1]												
(a)(ii)	<table border="1"><thead><tr><th>Temperature / °C</th><th>Height of foam / cm</th></tr></thead><tbody><tr><td>10</td><td>7.0</td></tr><tr><td>20</td><td>8.5</td></tr><tr><td>30</td><td>8.8</td></tr><tr><td>40</td><td>9.3</td></tr><tr><td>50</td><td>8.6</td></tr></tbody></table> Appropriate headings with units; Correct trend increasing below 40°C; Correct trend decreasing after 40°C; 1 dp;	Temperature / °C	Height of foam / cm	10	7.0	20	8.5	30	8.8	40	9.3	50	8.6	[4]
Temperature / °C	Height of foam / cm													
10	7.0													
20	8.5													
30	8.8													
40	9.3													
50	8.6													
(a)(iii)	40°C;	[1]												
(a)(iv)	Source of error: Temperature is not maintained during the five-minute incubation time of E+S mixture; Modification: Use a thermostatically-controlled water bath to ensure the temperature is constant throughout the incubation period;	[2]												

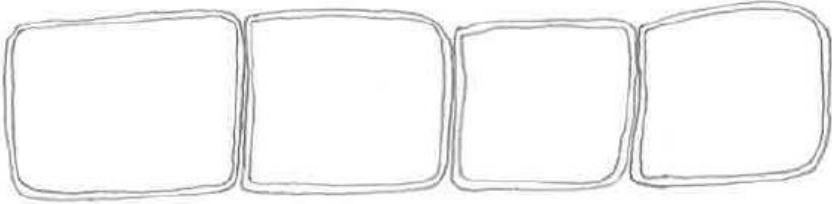
(a)(v)	Appropriate set up and/or description; State the measurement of oxygen production; Eg.  Perform a set up as shown in the figure to measure the amount of oxygen released with a gas syringe in cm³;;	[2]									
(b)(i)	<table border="1" data-bbox="320 1099 1310 1245"> <thead> <tr> <th data-bbox="320 1099 564 1137"></th><th colspan="2" data-bbox="564 1099 940 1137">Activity of catalase / arbitrary units</th></tr> <tr> <th data-bbox="320 1137 564 1176"></th><th data-bbox="564 1137 940 1176">extract A</th><th data-bbox="940 1137 1310 1176">extract B</th></tr> </thead> <tbody> <tr> <td data-bbox="320 1176 564 1245">Variance (s²)</td><td data-bbox="564 1176 940 1245"><u>0.36</u></td><td data-bbox="940 1176 1310 1245"><u>1.51</u></td></tr> </tbody> </table>		Activity of catalase / arbitrary units			extract A	extract B	Variance (s ²)	<u>0.36</u>	<u>1.51</u>	[1]
	Activity of catalase / arbitrary units										
	extract A	extract B									
Variance (s ²)	<u>0.36</u>	<u>1.51</u>									
(b)(ii)	$t = \frac{ \bar{x}_1 - \bar{x}_2 }{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$ $t = \frac{ 8.2 - 9.6 }{\sqrt{\left(\frac{0.36^2}{10} + \frac{1.51^2}{10}\right)}}$ $= 2.85 \text{ (2dp) ;}$ Number of degrees of freedom = 18 ;	[2]									
(b)(iii)	Reject H₀ (no marks on its own) Calculated t-value (2.85) > critical t-value (2.10) at P=0.05. 0.01 < P < 0.05 hence the difference between the activity of catalase from species A and B is significant;	[1]									

(c)(i)	As pH increases until optimum pH, height of foam increases + As pH increases beyond optimum pH, height of foam decreases; Ref. to change in residue R group charges; Eg. More H^+ present at lower pH results in basic amino acid residues in catalase being positively charged while fewer H^+ present at higher pH results in acidic residues being negatively charged. Ref. to disruption of ionic bonds AND loss of active site structure; Eg. Change in charges disrupts the ionic bonds required to stabilise the specific conformation of the enzyme and its active site. Ref. to reduced rate of enzyme-substrate complex formation hence reduced rate of product (O_2) formation;	[4]
(c)(ii)	Use different pH buffers with desired pH; OR Add varying volumes of acid and base to obtain the desired pH;	[1]
(d)(i)	Size of graph; Smooth and continuous best fit curve; Appropriate axes scale and label; Accurate plot;	[4]
(d)(ii)	As H_2O_2 concentration increases from 15-25%, activity of catalase increases slightly with decreasing rate from 235 to 247.5 AU, and plateaus; All the enzyme active sites are occupied and saturated, hence increase in substrate concentration does not increase rate of enzyme-substrate formation further; Substrate concentration is no longer the limiting factor;	[3]

(d)(iii)	 Activity of catalase / arbitrary units Concentration of H_2O_2 / % Showing line drawn from 6.5% H_2O_2 concentration + line drawn to Y-axis;	[1]
(d)(iv)	80 AU; Award marks according to value in graph	[1]

Question 2

(a)(i)	 Size of drawing at least $\frac{3}{4}$ of space given; Lines continuous and smooth; Accurate depiction of 3 layers within the vascular bundle; Accurate depiction of upper epidermis, lower epidermis and palisade mesophyll cell layer; Appropriate proportion of each layer;	[5]
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(a)(ii)	 Size of drawing at least $\frac{3}{4}$ of space given; Lines continuous and smooth; Accurately drew 4 cells with cell walls (with no other organelles); Appropriate proportion of cell wall thickness that is uniform;	[4]
(a)(iii)	Drawing length of chain of four cells = D cm (1dp) Magnification of drawing $= \times \frac{D \times 10^{-2} \text{ m}}{68 \times 10^{-6} \text{ m}}$ (2sf) Measurement of drawing length (D) in cm and 1dp; Show correct working of magnification ($\frac{D}{68}$) ; Show correct conversion of units; Final answer in 2-3 sf;	[4]
(a)(iv)	Fewer stomata present on the upper epidermis than lower epidermis;	[1]
(b)(i)	Colour of the hydrogencarbonate indicator;	[1]
(b)(ii)	Mass of pondweed placed into each beaker + use an electronic scale to weight a fixed mass of pondweed; Duration of exposure to each light intensity + use a timer to fix the duration; Heat emitted by the light source + use a heat filter; AVP variable + appropriate method to standardize;	[3]
(b)(iii)	Place the desk lamp at varying distance from the beaker, the further the distance, the lower the light intensity; Fix 3 distance for low, medium and high light intensities. Measure the distance with a plastic ruler. (a method that can be repeated);	[2]

(b)(iv)	<table><tr><th>Sample</th><th>Final colour of hydrogencarbonate indicator</th></tr><tr><td>P</td><td>Red</td></tr><tr><td>Q</td><td>Purple</td></tr><tr><td>R</td><td>Orange-red</td></tr></table> Appropriate headings; Correct final colour observed for all 3 samples;	Sample	Final colour of hydrogencarbonate indicator	P	Red	Q	Purple	R	Orange-red	[2]
Sample	Final colour of hydrogencarbonate indicator									
P	Red									
Q	Purple									
R	Orange-red									
(b)(v)	<table><tr><th>Sample</th><th>Light intensity</th></tr><tr><td>P</td><td>Medium</td></tr><tr><td>Q</td><td>High</td></tr><tr><td>R</td><td>Low</td></tr></table> Correct matching of light intensity to each sample;	Sample	Light intensity	P	Medium	Q	High	R	Low	[1]
Sample	Light intensity									
P	Medium									
Q	High									
R	Low									
(b)(vi)	The higher the light intensity, the higher the rate of photosynthesis, hence the higher the amount of CO₂ taken up; Ref. to CO₂ released in respiration;	[2]								
(b)(vii)	Expose a beaker of distilled water without pondweed to high light intensity and observe for colour change in the hydrogencarbonate indicator;	[1]								
(b)(viii)	Use a CO₂ meter to measure the concentration of CO₂ in the water; OR Use a pH meter to measure the change in pH of the water, instead of using hydrogencarbonate indicator;	[1]								